AI Incident Report Assistant — M2 Progress Report

Team Optimus Prime · Megathon X, Qualcomm Problem Statement 1 · Target: QIDK, Snapdragon 8 Gen 3 (SM8650, Hexagon v75) · Status as of repository commit `fcbee22`, 11 Oct 2026

1. Current approach and progress since M1

The core approach from M1 is unchanged: on-device Whisper transcribes; an on-device LLM only *proposes* field values as one JSON tool call per utterance; deterministic app code validates them, enforces the mandatory-field gate, asks follow-ups and files only after explicit confirmation; TTS speaks the replies; the form is a JSON schema. Since M1 the placeholder stages were replaced with Qualcomm runtimes, and two component choices changed:

Component	M1 plan	Current implementation	Status
LLM	Qwen3-4B-Instruct-2507, QAIRT/Genie bundle	Gemma-4-E2B-it, Q4_0 GGUF (3.04 GB), GenieX 0.7.1 llama.cpp runtime on the Hexagon NPU	Loads on the QIDK NPU; extraction on device not yet measured
ASR	Whisper, Voice AI ASR SDK	Whisper-Small, Voice AI ASR SDK 2.7.1.0, live partial captions	Runs on the QIDK; follow-up-turn fix awaiting device re-test
TTS	MeloTTS-EN	PiperTTS-EN via Voice AI TTS SDK 1.1.1.0 (22,050 Hz); Android TTS fallback	Integrated in code; not yet run on the QIDK
Form	Fetched from a proxy	Bundled 12-field form (7 required) loads offline; server optional	Implemented; host-tested

Why the LLM route changed. Qualcomm publishes Gemma-4-E2B only as a GenieX llama.cpp GGUF asset: there is no QAIRT bundle (a request for one was closed as not planned), and SM8650 is not on its chipset list. The GenieX AAR does ship Hexagon v75 kernels, so the model runs through GenieX's llama.cpp backend. Qwen3-4B via QAIRT remains the fallback.

Other progress in the app logic (pure Kotlin): the required-field loop, with read-back of unsure values and a confirmed "not specified" option; grouped optional questions; corrections; an `off_topic` tool that leaves the draft untouched; the reporter's name prefilled from a trusted device identity, never from speech; and submission safety (a model's `submit_report` is ignored, the gate is re-checked at submit, one idempotent report id per draft). The microphone now reopens when TTS reports completion, with a watchdog if that callback never arrives.

2. Experiments and results

E1–E3 concern the target hardware and its toolchain; E4 validates app logic on the host. All figures come from the project's own notes and logs.

Experiment	Configuration	Observation	Implication
E1. Deployment feasibility of Gemma-4-E2B on SM8650	Audit of AI Hub model manifests and release assets; contents of the GenieX 0.7.1 AAR	Single asset: Q4_0 GGUF for GenieX llama.cpp. No QAIRT bundle. SM8650 not listed by Qualcomm. AAR contains v73/v75/v79/v81 Hexagon kernels.	Feasible only via GenieX llama.cpp, unvalidated by Qualcomm on our chip
E2. LLM load on the QIDK	GenieX llama.cpp, context 4,096, compute unit NPU	The Hexagon session first failed to open (FastRPC error `0x80000406`: kernel not on the DSP search path); the GPU fallback took ~24 s to load. With the v75 kernel staged into the Whisper SDK's DSP directory, it loads on the NPU in ~4.3 s.	NPU path works on 8 Gen 3; ASR must initialise before the LLM
E3. Multi-turn speech recognition on the QIDK	Voice AI ASR SDK, one utterance per tap	The first utterance was transcribed, but every follow-up showed "listening" with no text: the SDK silently ignores `start()` while the previous session is open.	Fix committed (stop after each finished session; serialised start/stop); re-test pending
E4. Host pipeline tests with a scripted LLM	JUnit, `./gradlew :app:testDebugUnitTest`	Last recorded run: 94 of 94 tests passed, including the golden water-leak scenario (five fields filled, nothing invented) and ten acceptance utterances.	App logic verified; says nothing about Gemma's accuracy

No Qwen-versus-Gemma results are recorded, so no model comparison is reported. Gemma's extraction accuracy, per-turn latency and memory use on the QIDK are not yet measured.

3. Testing progress and current limitations

Category	Status	Strongest evidence
Component (host unit tests)	Done for app logic	94/94 passed at the last recorded run. The suite has since grown to 133 tests in 16 classes; the newer ones (off-topic, reporter prefill, hands-free reopen) have not been run for this report.
Integration (host, scripted LLM)	Done	End-to-end voice-loop test (extraction → gate → severity question → confirmation → report saved and posted once) plus 30 session tests.
Functional on device (gate, clarification, corrections, review)	Pending	Covered only by host tests so far.
Device bring-up	Partial	Whisper's first turn and Gemma's NPU load observed on the QIDK. Not yet verified: the spoken golden scenario, Piper TTS, and all three runtimes initialising in one APK.
Performance	Pending	Only model load time measured; per-turn latency, memory and airplane mode still to do.

Most important unresolved issue: the spoken multi-turn loop on the device. Silent Whisper follow-up turns (E3) and re-listening after TTS (now watchdog-backed) are fixed in code but not re-tested on the QIDK. Until they are, our design's latency target (median turn under 2.5 s) and on-device extraction quality remain unverified.

4. Plan for the remaining time

In dependency order, each with an observable check; no dates, as the remaining schedule is not documented.

1. **Close the voice loop on the device:** push the Piper model; confirm Whisper, Piper and GenieX initialise together; verify follow-ups transcribe and the mic reopens. *Done when* the golden scenario completes hands-free over at least three turns.
2. **Measure Gemma on the device:** speak the ten acceptance utterances; log per-field correctness, valid tool-call rate and per-turn timings. Malformed calls → enable GenieX's grammar constraint; Gemma unusable → switch to Qwen3-4B via QAIRT. *Done when* a results table exists.
3. **Functional tests on the device:** missing-field questions, “I don't know”, corrections (“actually 11 AM”), off-topic input, confirm-and-file. *Done when* each passes in a spoken session.
4. **Performance and offline behaviour:** memory (`dumpsys meminfo`), per-stage p50/p95 latency, full flow in airplane mode. *Done when* measured against the 2.5 s target.
5. **Regression and demo:** re-run all 133 host tests, fix regressions, record the demo and document measured numbers. If time allows, record M1's planned spoken test set for word error rate and field-level F1.

Evidence (project repository): `docs/LLM.md` for E1, E2 and E4; commit `d621c3c` for E3; `docs/DESIGN.md` §1.1 for the latency target; Qualcomm `ai-hub-models` issue #330 for the QAIRT-bundle request.